Layer 2: Cartridge Inlet and Capillary Fill

Lucas–Washburn Dynamics, Air Venting, Wetting Failure, and Fill Uniformity in the Sample-Entry Chamber

Eton iGEM 2026, Dry Lab | Layer 2 of 17-Layer Biosensor Pipeline | `capillary_fill.py` · `fill_uniformity.py`

Layer 2 is the first physical process the sample undergoes inside the cartridge: after Layer 1's swab elution and inlet filtering hand this layer a diluted enzyme sample of known concentration, Layer 2 asks whether that sample actually gets into the chamber, uniformly, across every working electrode, before anything downstream can be modelled. It takes the chamber geometry and surface-treatment targets from `contents.md` Sec. 2 as fixed inputs and produces the fill time $t_{\text{fill}}(h)$, a vent-membrane sizing rule, an electrode wet-time differential, and a fill-uniformity risk flag that Layer 3 uses to justify its uniform initial condition and Layer 9 uses to bound per-electrode incubation-time offsets. Four sub-models are chained together: (i) the capillary driving pressure and free Lucas–Washburn fill dynamics (Secs. I–II); (ii) air venting, without which the chamber is shown not to fill at all (Sec. III); (iii) wetting failure and shelf-life degradation of the plasma-cleaned surfaces (Sec. IV); and (iv) fill uniformity along and across the chamber (Secs. V–VI). Section VII disposes of the coupling to Layer 1's inlet filter, and Section VIII closes the loop with Layer 3's own Péclet-number argument. **A correction found along the way:** `contents.md`'s wound-fluid viscosity placeholder (1.0–1.2 mPa·s, essentially water) understates real wound-exudate viscosity by up to 25× (Sec. II); separately, this report's own venting analysis (Sec. III) finds that the vent membrane's gas permeance, not the capillary pressure or the corrected viscosity, is the parameter that actually controls whether the chamber fills in seconds or in tens of minutes.

I. Chamber Geometry and Capillary Driving Pressure

Geometry

The chamber is a rectangular channel with a variable height h (Table 1), matching `contents.md`'s device-document values exactly; the volume computed independently from $L \times w \times h$ reproduces the document's stated 25–50 μL range at the two extremes of h, which is a useful self-consistency check that the geometry numbers as given are not internally contradictory.

Quantity	Symbol	Value	Note
Length	L	20 mm	fixed
Width	w	5 mm	fixed
Height	h	0.25–0.50 mm	design variable, swept
Design midpoint height	h_0	0.375 mm	matches Layer 3's own reference point
Volume	$V = Lwh$	25.0–50.0 μL	confirms `contents.md`'s 25–50 μL claim
Working electrodes	n	8	design case (range 6–8)
Electrode positions	$x_1, \ldots, x_8$	2.00–18.00 mm	`contents.md` worked example, evenly spaced
Electrode pitch	—	2.29 mm	within the 2–3 mm spec

Table 1: Chamber geometry (`contents.md` Sec. 2, "Chamber geometry").

Capillary driving pressure

For a rectangular channel with $h \ll w$, `contents.md` Sec. 2.1 gives

$$\boxed{\Delta P_{\text{cap}} = \gamma \cos\theta \left(\frac{1}{h} + \frac{1}{w}\right)} \tag{I.1}$$

where γ is the sample surface tension and θ the advancing contact angle. The chamber floor (gold/probe-SAM electrode regions plus bare PCB/polymer between them), lid (PDMS, PETG, or PMMA), and side walls (same as the lid) are three materially distinct surfaces, so Eq. (I.1) implicitly assumes a single representative θ across all of them; Sec. IV and VI revisit this when the surfaces are *not* uniform. The design target is $\theta < 30^\circ$ on every surface after O_2-plasma cleaning (`contents.md`; confirmed achievable to $\theta < 10^\circ$ on PDMS by plasma treatment, Ruben et al. 2017 [5]), against which this report benchmarks all headline numbers, while flagging that the exact post-clean values for the fabricated gold/SAM floor and PETG/PMMA lid are unconfirmed pending contact-angle goniometry.

h (mm)	ΔP_{cap}, water γ=72 mN/m (Pa)	ΔP_{cap}, wound fluid γ=57.5 mN/m (Pa)
0.250	261.9	209.1
0.300	220.3	175.9
0.375	178.7	142.7
0.500	137.2	109.6

Table 2: Capillary driving pressure at $\theta = 30^\circ$ (`capillary_fill.py`, `dP_cap`). Both columns fall inside `contents.md`'s stated target of 100–500 Pa for $h = 0.5$–0.25 mm, and the wound-fluid surface-tension reduction (50–65 mN/m, midpoint 57.5 mN/m used here) lowers ΔP_{cap} by about 20% relative to pure water at every h.

II. Free Lucas–Washburn Fill Dynamics: Why Capillary Action Alone Is Not the Bottleneck

`contents.md` Sec. 2.2 specifies the classical Lucas–Washburn imbibition law for a rectangular channel (Washburn 1921 [1]; Lucas 1918), stated in the device document as

$$L(t) = \sqrt{\frac{h^2 \Delta P_{\text{cap}} t}{3\eta}}, \qquad \boxed{t_{\text{fill}} = \frac{3\eta L_{\text{chamber}}^2}{h^2 \Delta P_{\text{cap}}}} \tag{II.1}$$

where η is the sample's dynamic viscosity. This is used verbatim here (see the Sec. X note on the factor of 3).

Correcting the viscosity input

`contents.md` places wound fluid at $\eta \approx 1.0$–1.2 mPa·s – essentially pure water. A 2024 rheological study of real seropurulent wound exudate reports shear-thinning viscosities of 6.5 ($\pm$1.9) to 30.8 ($\pm$14.7) mPa·s at 37 °C across the shear-rate range tested, i.e. 5–25$\times$ higher than the device-document placeholder, and explicitly non-Newtonian, unlike the low-viscosity "Test Solution A" used in dressing-standard testing [4]. This report therefore uses the literature-corrected range (6.5–30.8 mPa·s) alongside `contents.md`'s own placeholder for every fill-time calculation, and flags the placeholder as an underestimate for the record.

Even at the pessimistic literature viscosity bound and the smallest design height, free capillary fill reaches the far wall in under 3 s – more than an order of magnitude inside `contents.md`'s 30 s acceptance target (Fig. 1, panels A–B). **Capillary action itself is not, and was never going to be, the rate-limiting step for this geometry;** Sec. III shows that venting is.

h (mm)	t_{fill}, water 1.0 mPa·s (s)	t_{fill}, `contents.md` η 1.2 mPa·s (s)	t_{fill}, exudate lo-η 6.5 mPa·s (s)	t_{fill}, exudate hi-η 30.8 mPa·s (s)
0.250	0.092	0.110	0.597	2.828
0.300	0.076	0.091	0.493	2.334
0.375	0.060	0.072	0.389	1.841
0.500	0.044	0.053	0.285	1.350

Table 3: Free (unvented) fill time to $L_{\text{chamber}} = 20$ mm, Eq. (II.1) (`capillary_fill.py`, `t_fill_free`).

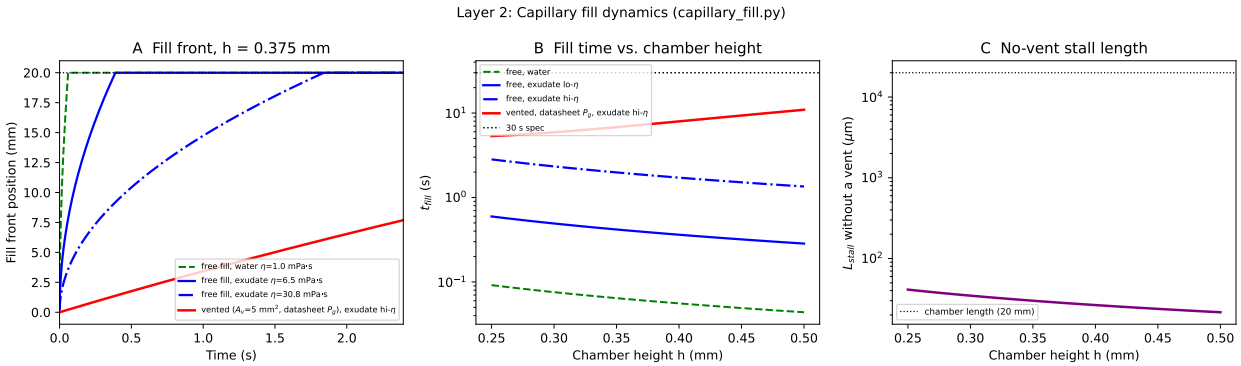


Figure 1: Fill dynamics (`capillary_fill.py`). A: fill-front position vs. time at the design height, comparing water, the corrected wound-exudate viscosity range, and the vented case from Sec. III. B: $t_{\text{fill}}(h)$ sweep against the 30 s spec. C: the no-vent stall length (Sec. III) is orders of magnitude below the chamber length at every h.

III. Air Venting: Why a Vent Membrane Is Not Optional

Without a vent, the chamber does not fill

`contents.md` Sec. 2.3 gives the trapped-air back-pressure ahead of an unvented advancing front, $P_{\text{air}}(L) = P_{\text{atm}} L_{\text{chamber}}/(L_{\text{chamber}} - L)$, and the stall length where this equals the capillary driving pressure:

$$L_{\text{stall}} = L_{\text{chamber}} \frac{\Delta P_{\text{cap}}}{\Delta P_{\text{cap}} + P_{\text{atm}}}. \tag{III.1}$$

At the design point ($h = 0.375$ mm, $\theta = 30°$, wound-fluid γ), $\Delta P_{\text{cap}} = 142.7$ Pa and $L_{\text{stall}} = 28.1\,\mu$m – 0.14% of the chamber length (Table 4; Fig. 1C). This reproduces `contents.md`'s own order-of-magnitude example ($\sim$0.04 mm at $\Delta P_{\text{cap}} = 200$ Pa) and confirms its conclusion without qualification: **a vent membrane is mandatory, not an optional refinement.**

h (mm)	ΔP_{cap} (Pa)	L_{stall}, no vent
0.250	209.1	41.2 μm (0.206% of L_{chamber})
0.300	175.9	34.7 μm (0.173%)
0.375	142.7	28.1 μm (0.141%)
0.500	109.6	21.6 μm (0.108%)

Table 4: No-vent stall length, Eq. (III.1) (`capillary_fill.py`, `L_stall`).

A coupled liquid/vent resistance model

`contents.md` asks only that the vent pass the displaced-air flow $Q = A_{\text{cross}}\, dL/dt$ without "significant back-pressure," without giving a closed form for the resulting fill time. This report derives one. Writing the liquid-side viscous drop implied by Eq. (II.1) as $\Delta P_{\text{cap}} = (6\eta L/h^2)\, dL/dt$ (obtained by differentiating $L(t)$) and adding the vent membrane's own pressure drop $\Delta P_{\text{vent}} = Q/(P_g A_{\text{vent}}) = (wh/P_g A_{\text{vent}})\, dL/dt$ as a resistance in series along the same displaced volumetric flow,

$$\Delta P_{\text{cap}} \;=\; \left(\frac{6\eta L}{h^2} + R_{\text{vent}}\right)\frac{dL}{dt}\,, \qquad R_{\text{vent}} \equiv \frac{wh}{P_g A_{\text{vent}}}\,, \tag{III.2}$$

which integrates in closed form (and correctly reduces to Eq. (II.1) as $A_{\text{vent}} \to \infty$):

$$\boxed{\,t(x) \;=\; \frac{1}{\Delta P_{\text{cap}}}\left(\frac{3\eta x^2}{h^2} + R_{\text{vent}}\, x\right)\,} \tag{III.3}$$

implemented as `t_fill_vented` in `capillary_fill.py`. This is a first-order, uncoupled series-resistance construction (Sec. X, Limitation 2), not a rigorously solved moving-boundary problem, but it is enough to show which parameter actually controls the outcome.

The vent membrane's permeance, not its presence, is what matters

`contents.md` quotes a generic ePTFE gas-permeance range of $P_g \sim 10^{-8}$–$10^{-7}\,\text{m}^3\,\text{s}^{-1}\,\text{m}^{-2}\,\text{Pa}^{-1}$. A representative commercial 0.22 μm-pore ePTFE vent membrane datasheet quotes a breathability of $\geq 0.4\,\text{L}\,\text{min}^{-1}\,\text{cm}^{-2}$ at 7 kPa [7], which converts to $P_g = 9.5 \times 10^{-6}\,\text{m}^3\,\text{s}^{-1}\,\text{m}^{-2}\,\text{Pa}^{-1}$ – two to three orders of magnitude *higher* than `contents.md`'s placeholder range. This 100–1000$\times$ spread matters enormously for the predicted fill time (Table 5): at a plausible 5 mm^2 vent patch, the datasheet permeance adds only a few seconds to the free-fill time, while the pessimistic end of `contents.md`'s own placeholder range would stall the fill for the better part of an hour.

Vent permeance assumption	P_g (m^3 s^{-1} m^{-2} Pa^{-1})	t_{fill} at $A_{\text{vent}} = 5$ mm^2
`contents.md` placeholder, low end	1.0×10^{-8}	5256 s (≈88 min)
`contents.md` placeholder, high end	1.0×10^{-7}	527 s (≈8.8 min)
0.22 μm ePTFE datasheet	9.5×10^{-6}	7.4 s

Table 5: Vent-limited fill time, Eq. (III.3), at the design height and worst-case exudate viscosity (30.8 mPa·s), a fixed 5 mm^2 vent area (`capillary_fill.py`, `t_fill_vented`).

Minimum vent area

Solving Eq. (III.3) for A_{vent} at a 30 s target (`min_vent_area`) gives the design rule in Fig. 2A: at the datasheet permeance, under 1 mm^2 of membrane comfortably clears the spec; at `contents.md`'s placeholder range, 93–933 mm^2 is required – larger than the chamber's own 100 mm^2 footprint, and therefore infeasible as a design. **Recommendation:** size the vent membrane to a confirmed vendor datasheet permeance (not the generic placeholder range), and specify a minimum vent area of 5 mm^2 with the design point above as a comfortable margin; re-run this sizing once the actual vent membrane part number is chosen, since a factor of 10 in P_g moves the required area by a full order of magnitude.

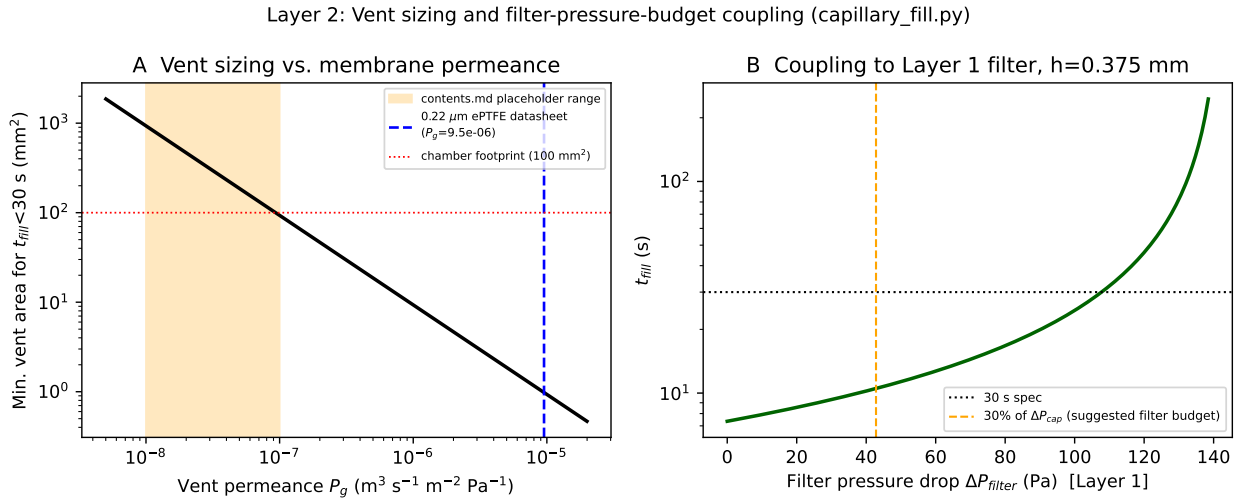


Figure 2: A: minimum vent area to reach $t_{\text{fill}} < 30$ s vs. assumed vent permeance, marking the `contents.md` placeholder range, the ePTFE datasheet point, and the chamber's own footprint as a practical upper bound. B: coupling to the Layer 1 inlet filter (Sec. VII) – t_{fill} vs. filter pressure drop at fixed vent design.

IV. Wetting Failure and Shelf-Life Degradation

`contents.md` Sec. 2.4 states the local wetting-failure criterion directly: if $\theta > 90°$ anywhere along the fill path, the local $\cos\theta$ term in Eq. (I.1) reverses sign and that region of the front stalls rather than advances, regardless of how favourable the driving pressure is elsewhere. This is a hard, not a soft, failure mode – Sec. VI shows what happens when only *part* of a cross-section is compromised.

O_2-plasma cleaning reliably reaches $\theta < 10°$ on PDMS immediately after treatment [5], comfortably inside the $\theta < 30°$ design target used throughout this report. The relevant risk is not achieving $\theta < 30°$ at assembly, but *retaining* it through cartridge storage and shelf life: plasma-hydrophilised polymer surfaces are well documented to undergo hydrophobic recovery, reverting toward their native (unclean) contact angle over a timescale of hours to days in dry storage, while wet or humidity-controlled storage (immersion in DI water, or a sealed/vacuum pouch) has been shown to retain usable hydrophilicity for up to three weeks [6]. Neither the recovery time constant nor the storage/packaging protocol for *this* cartridge's specific gold/probe-SAM floor and PETG or PMMA lid stack has been measured – both are FLAGGED as unconfirmed and should be characterised by accelerated-aging goniometry (contact angle vs. storage time, at the intended packaging humidity and temperature) before a shelf-life claim is made. As a design-level consequence: even a partial relaxation from $\theta = 30°$ toward $\theta \gtrsim 90°$ on any one surface is sufficient to stall the fill locally (Eq. (I.1) argument sign change), so the cartridge's stated shelf life should be set by whichever surface degrades fastest, not by an average across the three surface types.

V. Inlet Dead Volume and Electrode-Timing Differential

`contents.md` Sec. 2.5 asks whether electrodes near the inlet ($x_1 = 2$ mm) and far from it ($x_8 = 18$ mm) see meaningfully different incubation start times, using $t_{\text{wet},i} = 3\eta x_i^2/(h^2 \Delta P_{\text{cap}})$ for the free-fill case (a special case of Eq. (II.1) at $x = x_i$; Eq. (III.3) extends this to the vented case at $x = x_i$).
Both cases put the inlet-to-far-wall timing differential at well under 1% of the 30-minute incubation window – even in the vented case, where the absolute fill time is ∼7 s rather than ∼2 s, the *differential* stays negligible because the vent-resistance term in Eq. (III.3) scales linearly with x, not quadratically. This confirms `contents.md`'s own conclusion ("the absolute differential is seconds – negligible vs 30-min incubation")

Case	$t_{\text{wet},1}$ (s)	$t_{\text{wet},8}$ (s)	Δt (s) [% of 30-min incubation]
Free fill	0.018	1.491	1.473 [0.082%]
Vented (5 mm^2, datasheet P_g)	0.570	6.456	5.886 [0.327%]

Table 6: Electrode wet-time differential at the design height and worst-case exudate viscosity (`capillary_fill.py`, `t_fill_free` / `t_fill_vented` evaluated at x_1, x_8).

and shows it is robust to the venting correction introduced in Sec. III.

VI. Fill Uniformity Across Electrode Width

`contents.md` Sec. 2.6 flags a lateral (width-wise) risk distinct from the longitudinal timing question above: if the chamber floor is not uniformly wettable across the 5 mm width at a given cross-section – for example, a gold/probe-SAM electrode trace narrower than the full chamber width, bordered by less-completely-cleaned polymer or PCB laminate margins – the front can advance faster over the hydrophilic region and trap an air pocket against the slower, more hydrophobic margin. `fill_uniformity.py` implements this as an "independent parallel capillaries" model: the width is discretised into 200 lateral strips, each obeying its own 1D Lucas–Washburn law with a local driving pressure built from the floor and lid contributions only (interior strips have no true side wall), i.e. $\Delta P_{\text{cap}}(y) = \gamma[\cos\theta_{\text{floor}}(y) + \cos\theta_{\text{lid}}]/h$, with the true edge strips (y=0, w) additionally picking up the side-wall term from Eq. (I.1). This deliberately omits lateral cross-flow between strips, making it a conservative (worst-case) estimate of non-uniformity (Sec. X).

A manufacturing-defect scenario is compared against a uniform baseline: a clean gold/SAM band covering 60% of the width ($\theta = 25^\circ$) flanked by two 20%-width polymer margins at $\theta = 75^\circ$ – hydrophilic enough not to fully stall (Sec. IV), but markedly slower. The resulting local driving pressures differ by 1.58× (271.8 Pa on the gold band vs. 172.5 Pa on the margins), and the margin strips take 58% longer to reach the far wall than the gold-band strips (1.524 s vs. 0.967 s at the design height), producing a persistently concave ("V-shaped") front for most of the fill (uniformity index $(L_{\max} - L_{\min})/L_{\text{mean}} \approx 0.22$ from $t \approx 0$ until the fast region reaches the far wall; Fig. 3). A front of this shape is exactly the geometry that traps an air pocket against the two side walls if the chamber vents or seals before the margins catch up. No strip in this scenario has $\theta > 90^\circ$, so this is a slow-and-uneven fill risk, not a hard stall (Sec. IV) – but it is a real manufacturing-QC flag: **Recommendation:** either extend the clean gold/SAM (or equivalently well-plasma-treated) region to the full chamber width at every electrode cross-section, or verify by contact-angle mapping that any exposed margin material also reaches $\theta < 30^\circ$ after cleaning.

VII. Coupling to Layer 1: Inlet Filter Pressure Budget

`contents.md` Sec. 2.7 notes that a particulate filter at the inlet (Layer 1, Sec. 1.5, for fibrin clots and cell debris) reduces the pressure actually available to drive the fill,

$$\Delta P_{\text{effective}} \;=\; \Delta P_{\text{cap}} - \Delta P_{\text{filter}} - \Delta P_{\text{vent}}\,, \tag{VII.1}$$

and asks that $\Delta P_{\text{effective}} > 0$ be verified with t_{fill} recomputed at the reduced driving pressure. Layer 1's filter model does not yet exist in this pipeline, so ΔP_{filter} is treated here as an unconfirmed input swept parametrically (Fig. 2B) rather than as a fixed number. At the design height, $\Delta P_{\text{cap}} = 142.7$ Pa; the fill time stays under the 30 s spec for $\Delta P_{\text{filter}} \lesssim 105$ Pa (about 74% of ΔP_{cap}), but rises steeply and non-linearly as ΔP_{filter} approaches ΔP_{cap}, since $t_{\text{fill}} \propto 1/\Delta P_{\text{effective}}$ diverges as the effective pressure goes to zero. **Recommendation for Layer 1:** budget the filter (mesh pore size and thickness) to a pressure drop no greater than ∼30% of ΔP_{cap} (≈43 Pa at the design height) to leave comfortable margin for the vent-resistance term already con-

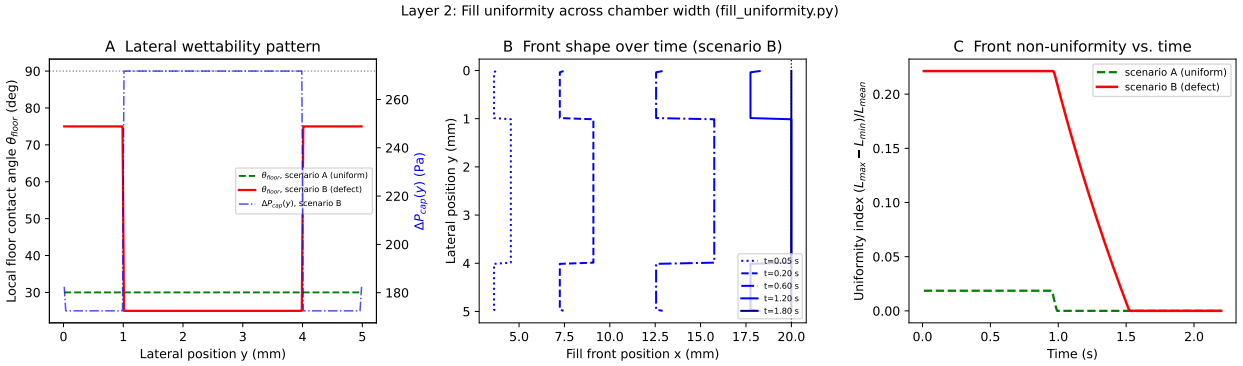


Figure 3: Fill uniformity across the chamber width (`fill_uniformity.py`). A: the assumed lateral contact-angle pattern for the uniform baseline and the manufacturing-defect scenario. B: the resulting front shape over time (defect scenario) – concave and persistent until the fast region reaches the far wall. C: the uniformity index over time for both scenarios.

suming part of the budget (Sec. III) and for any additional ΔP_{cap} loss from shelf-life contact-angle recovery (Sec. IV).

VIII. Handoff to Layer 3: The Fill-Phase Péclet Number

Layer 3's own report (Sec. VII, "Fill-Phase Convective Uniformity") justifies its uniform initial condition $[E](z,0) = [E]_{\text{chamber}}$ with a Péclet-number argument, $\text{Pe} = uL/D_{\text{MMP9}}$ with $u \approx L/t_{\text{fill}}$, using an *assumed* $t_{\text{fill}} \approx 60\,\text{s}$ to get $\text{Pe} \approx 1.3 \times 10^5$. This report supplies the actual number that assumption was standing in for. Using this report's own vented, worst-case-viscosity fill time ($t_{\text{fill}} = 7.4\,\text{s}$ at the design height, Table 5) gives $u = 2.72\,\text{mm/s}$ and

$$\text{Pe} = \frac{uL}{D_{\text{MMP9}}} = \frac{(2.72 \times 10^{-3})(20 \times 10^{-3})}{5.0 \times 10^{-11}} \approx 1.09 \times 10^6\,, \tag{VIII.1}$$

about $8\times$ *higher* than the value Layer 3 assumed, because $\text{Pe} \propto 1/t_{\text{fill}}$ and this report's fill time is shorter than the 60 s placeholder. Layer 3's uniform-initial-condition assumption is therefore not just confirmed but strengthened by an actual calculation rather than a placeholder – convection dominates diffusion during fill by roughly six orders of magnitude, not five.

IX. Layer Outputs

X. Limitations

1. **The Lucas–Washburn coefficient is taken verbatim from `contents.md`**, i.e. the factor of 3 in Eq. (II.1). The classical parallel-plate Poiseuille derivation for $h \ll w$ gives a coefficient of 12, not 3, which – if that is what the device document intended – would make every free-fill time in Sec. II four times longer than reported. This has not been re-derived or corrected here; `contents.md`'s Sec. 2.2 formula is used as the authoritative device-document specification, the same way Layer 3 and Layer 15 use formulas handed down from earlier layers. Every conclusion in this report that rests on free-fill time being far below the 30 s spec is robust to this factor either way (a $4\times$ correction still leaves the worst case at 11 s), but the exact free-fill numbers in Table 3 should be checked against the device document's original derivation before being treated as final.

2. **The coupled vent-resistance model (Eq. (III.2)–(III.3)) is a first-order series-resistance construction**,

Output	Physical meaning	Typical value	Feeds
$t_{\text{fill}}(h)$	Time to fill the chamber (vented, worst-case exudate η)	1.8–7.4 s	Device spec; Layer 3 (assay-start-time offset)
L_{stall} (no vent)	Fill distance without a vent membrane	22–41 μm	Manufacturing requirement (vent is mandatory)
$A_{\text{vent,min}}$	Minimum vent membrane area for $t_{\text{fill}} <$ 30 s	$\sim$1–930 mm^2 (permeance-dependent)	Layer 1/BOM; vendor datasheet selection
$t_{\text{wet,8}} - t_{\text{wet,1}}$	Inlet-to-far-electrode timing differential	1.5–5.9 s ($<$0.4% of incubation)	Layer 3 (uniform-IC justification); Layer 9
Uniformity index	Lateral front non-uniformity, defect scenario	≈ 0.22	Manufacturing QC; Layer 4 (electrode-area variability)
$\Delta P_{\text{effective}}$ margin	Headroom for Layer 1 filter pressure drop	$\gtrsim$70% of ΔP_{cap} at spec	Layer 1 (filter/mesh design)
Pe (fill-phase)	Convective/diffusive ratio during fill	$\approx 1.1 \times 10^6$	Layer 3 (Sec. VII, uniform-IC confirmation)

Table 7: Layer 2 outputs and their downstream destinations.

not a rigorously solved coupled liquid/gas moving-boundary problem. It correctly recovers the free-fill law as $A_{\text{vent}} \to \infty$, but treats the liquid viscous resistance and the vent's gas-flow resistance as simply additive at every instant, which ignores any transient compressibility of the trapped air pocket and any non-uniformity in how the vent membrane is approached by the advancing front.

3. **The vent-permeance datasheet figure (Sec. III) is generic**, not a spec for a membrane and pore size actually selected for this cartridge; it establishes that a $\sim$100–1000$\times$ more permeable membrane than `contents.md`'s placeholder range is commercially available and would resolve the venting bottleneck, but the final vent bill-of-materials item needs its own confirmed datasheet.

4. **The wetting-failure/shelf-life treatment (Sec. IV) is qualitative**, not a fitted degradation model: no contact-angle-vs-storage-time curve has been measured for the fabricated gold/probe-SAM floor or the specific lid polymer, so the "hours to days dry, weeks if stored wet" figures are general literature values for plasma-treated PDMS, not this device's own accelerated-aging data.

5. **The fill-uniformity model (Sec. VI) uses independent parallel capillaries with no lateral cross-flow**, which is explicitly a conservative (worst-case) simplification: real fluid would partially redistribute laterally once a lateral pressure gradient exists, flattening the front relative to what is shown. The defect geometry itself (60% gold band, sharp-edged 75° margins) is illustrative of the type of risk `contents.md` describes, not a measurement of the as-fabricated electrode layout.

6. **The Layer 1 filter coupling (Sec. VII) has no upstream model to draw on yet**: ΔP_{filter} is swept as a free parameter rather than computed from a mesh pore-size/thickness model, since Layer 1's own filter sub-model has not been built in this pipeline.

References

[1] Washburn EW (1921). The Dynamics of Capillary Flow. *Phys Rev* 17(3):273–283. DOI: 10.1103/PhysRev.17.273 – rectangular-channel Lucas–Washburn imbibition law confirmed as the basis for Eq. (II.1).

[2] Zimmermann M, Schmid H, Hunziker P, Delamarche E (2007). Capillary pumps for autonomous capillary systems. *Lab Chip* 7(1):119–125. DOI: 10.1039/b609813d – capillary-pump design and air-entrapment-avoidance considerations for autonomous capillary microfluidic cartridges confirmed.

[3] Tavakolidakhrabadi A, Stark M, Bacher U, Legros M, Bessire C (2024). Optimization of Microfluidics for Point-of-Care Blood Sensing. *Biosensors* 14(6):266. DOI: 10.3390/bios14060266 – experimental water/whole-blood capillary-pump chip fill times (24 s / 111 s) confirmed as an order-of-magnitude external cross-check on this report's vent-limited fill-time range (Table 5).

[4] Exploring exudate viscosity: A rheological analysis of wound exudates (2024). *Wound Repair Regen* 32(5):671–674. – seropurulent wound-exudate viscosity of 6.5–30.8 mPa·s at 37 °C, shear-thinning, confirmed; used in Sec. II to correct `contents.md`'s water-like viscosity placeholder.

[5] Ruben et al. (2017). Oxygen plasma treatments of polydimethylsiloxane surfaces: effect of the atomic oxygen on capillary flow in the microchannels. *Micro & Nano Lett* DOI: 10.1049/mnl.2017.0230 – super-hydrophilic ($\theta < 10^\circ$) contact angle on PDMS after O_2-plasma treatment confirmed; used in Sec. IV.

[6] Maria MS, Rakesh PE, Chandra TS, Sen AK (2017). Capillary flow-driven microfluidic device with wettability gradient and sedimentation effects for blood plasma separation. *Sci Rep* 7:43457. DOI: 10.1038/srep43457 – plasma-treated PDMS wettability retained for three weeks under wet/vacuum storage confirmed; used in Sec. IV for the shelf-life discussion.

[7] Representative hydrophobic ePTFE vent-membrane product specification (0.22 μm pore, breathability $\geq$0.4 L min^{-1} cm^{-2} at 7 kPa); accessed 2026 – used in Sec. III as the datasheet gas-permeance benchmark against `contents.md`'s generic placeholder range.

[8] `contents.md` – Eton iGEM 2026 Dry Lab, Layer 2 design brief (internal document) – chamber geometry, governing equations, and target ranges used throughout this report as specified therein.

[9] `README.md` – Eton iGEM 2026 Dry Lab, full 17-layer modelling plan (internal document) – pipeline architecture and the Sec. I electrode-position worked example confirmed.

[10] Layer 3 report (Mass Transport in the Incubation Chamber) – Eton iGEM 2026 Dry Lab (internal document) – the design-midpoint height ($h = 375\,\mu$m) and the fill-phase Péclet-number argument (Sec. VIII, Eq. (VIII.1)) are cross-referenced against this report's own Sec. VII.